FACULTY OF COMPUTING

SEMESTER 2 2025/2026

SECP3843 SPECIAL TOPIC IN DATA ENGINEERING

SECTION 01

DATABASE AND SYSTEM DESIGN FOR MONITORING

DASHBOARD TUTORIAL

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GROUP 5

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1. Main Problem

PT.MPM is a packaging producer in Indonesia. They face challenges in effectively assessing and monitoring company's performance.

The organisation receives sales forecasts in multiple formats (annual, quarterly, monthly) and performs aggregate and material planning based on production levels, historical sales data, inventory availability and managerial insights. However, due to the lack of suitable applications and systems, these tasks are time-consuming and inefficient. Resulting in ineffective decision making, thus, reducing its competitiveness in the market.

2. Main Solution

To address these challenges, a business intelligence solution was developed in the form of two dashboards using Google Data Studio. The two dashboards are a Sales Performance Dashboard and a Production Performance Dashboard respectively and are fed by data warehouses created in Google Sheets.

The dashboards allows staff and management to

- Monitor sales and production performance against KPIs
- Compare projected versus actual outcomes
- Assess resource utilisation and production efficiency
- Support aggregate and material planning

3. User Story

This section outlines the user's goals and expectations for the system.

3.1 US01 - Monitor Sales Performance

As a senior management user,
I want to view sales KPIs such as revenue, forecast revenue, forecast sales and total sales across products, product categories and customers
So that I can evaluate business performance by assessing whether sales performance meets the company's KPIs and identify performance gaps.

3.2 US02 - Monitor Production Performance

As a senior management user,
I want to view production KPIs such as planned production, actual production, OEE, availability, performance and quality across machines and production divisions
So that I can evaluate manufacturing performance and identify inefficiencies in the production process.

3.3 US03 - Compare Planned vs Actual Performance

As a senior management user,
I want to compare planned values against actual values for both sales and production
So that I can identify deviations to improve decision-making and operational planning.

4. Entity Relationship Diagram (ERD)

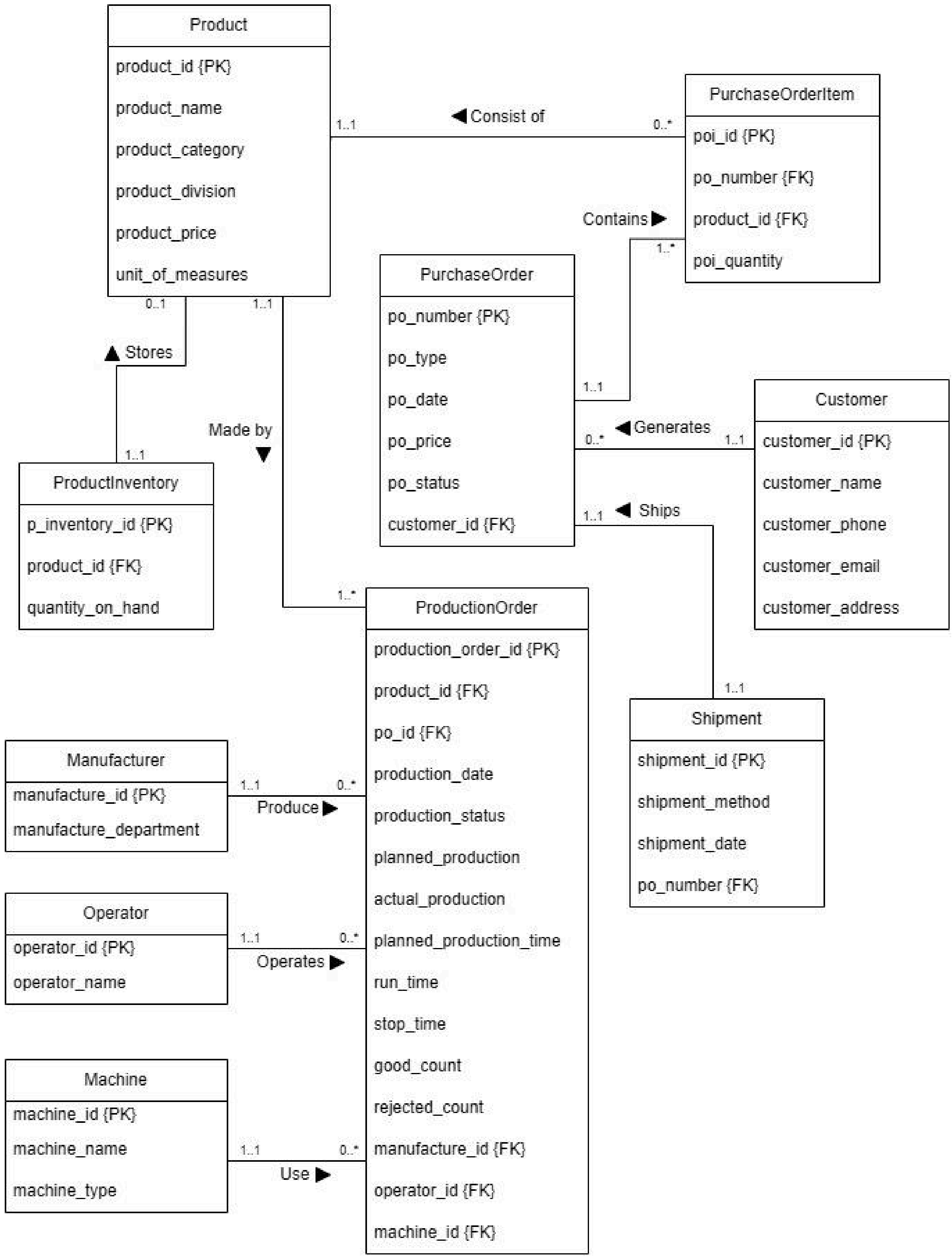

Figure 1 Entity Relationship Diagram

5. Use Case Diagram

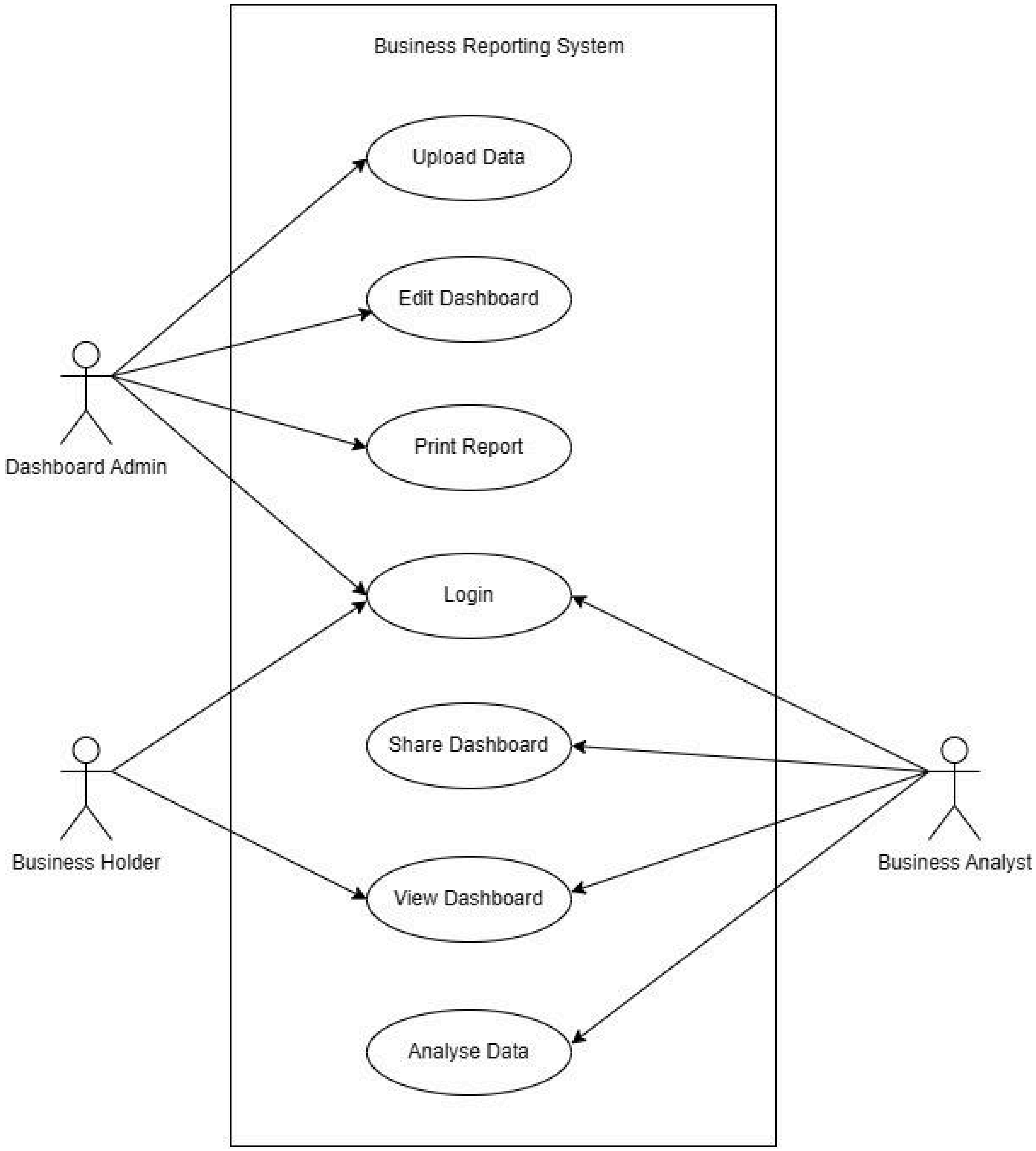

Figure 2 Use Case Diagram

6. Data Architecture

The following figure is the data architecture involved in the system.

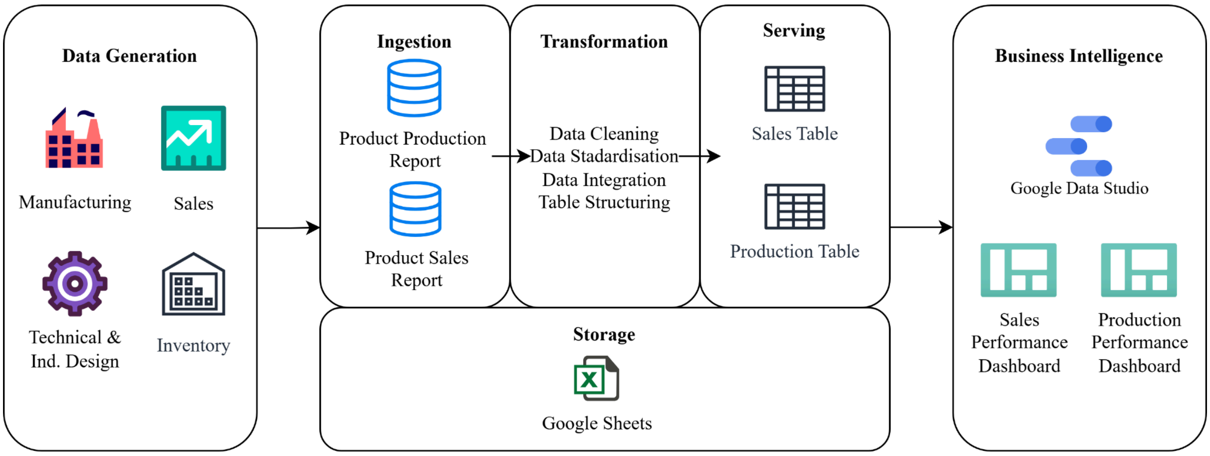

Figure 3 Data Architecture

Based on the data flow diagram in the conceptual model presented by the writer, there are four entities producing data, including Technical & Ind. Design, Inventory & Warehouse, Manufacturing and Sales. The database includes Product Information, Product Stock Information, Production Information and Sales Information.

In the dashboard production, only the product production report and product sales report were ingested from the Production Information and Sales Information, respectively. All data was converted to Google Sheets format.

Several operations were performed during the transformation phase, including the removal of irrelevant data, standardisation of data types, and table structuring. A data warehouse, comprising a sales table and a production performance table, was created in Google Sheets.

The Google Sheets were then uploaded to Google Data Studio for dashboard creation.

7. Star Schema

The data warehouse for PT. MPM was designed using the star schema approach. Two separate schemas were developed: a Sales Star Schema and a Production Star Schema to support two main analytical areas of the organization, which are sales performance and production performance.

7.1 Sales Star Schema

The Sales Star Schema was designed to support sales performance analysis. The central fact table, fact_sales stores quantitative sales measures such as Revenue, Forecast Revenue, Forecast Sales, PO Regular, PO Extra, Price and Total Sales for the company to evaluate sales performance and monitor forecasting accuracy.

The fact table is connected to three dimension tables (as shown in Figure 4):

- dim_customer: contains customer-related information
- dim_product: stores descriptive product attributes
- dim_date: stores time-related attributes

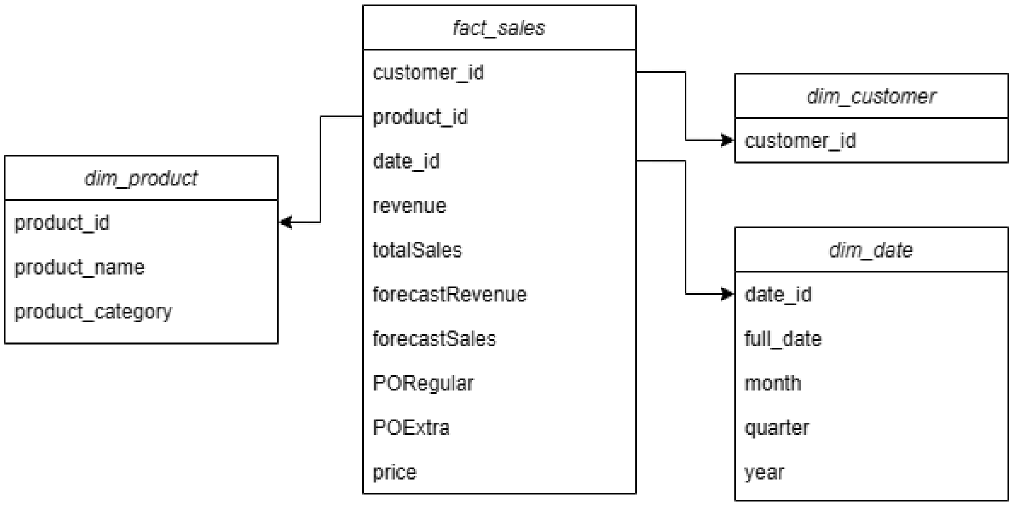

Figure 4 Sales Star Schema

The Sales Star Schema enables users to analyze revenue, sales and forecast performance across different customer, product categories and time periods.

7.2 Production Star Schema

The Production Star Schema was designed to support production performance monitoring and operational analysis. The central fact table, Fact_Production, contains production-related measures such as Planned Production, Actual Production, Planned Production Time, Good Count, Reject Count, Run Time, Stop Time, Availability, Performance, Quality and Overall Equipment Effectiveness (OEE).

The fact table is connected to four dimension tables (as shown in Figure 5):

- dim_machine: contains machine-related information used in production
- dim_operator: stores operator information responsible for production activities
- dim_product_division: contains production division information
- dim_date: stores time-related attributes

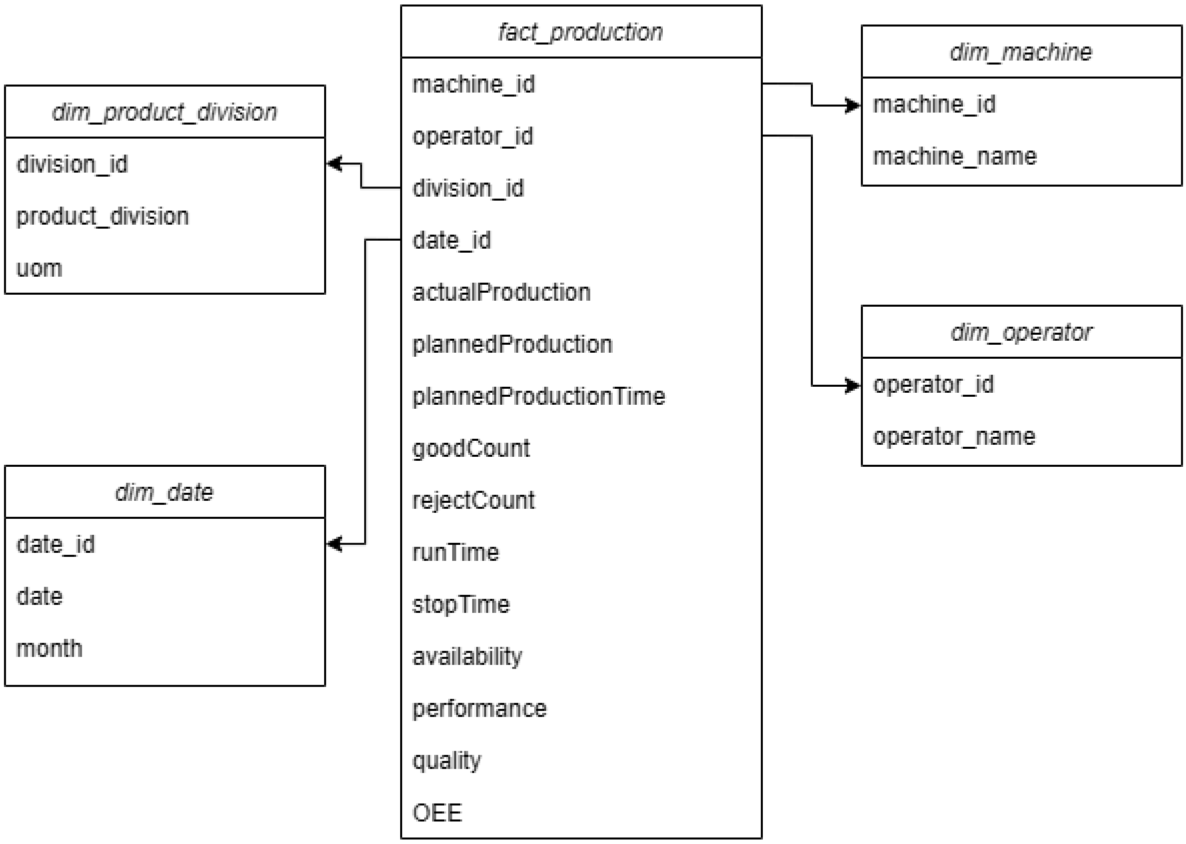

Figure 5 Production Star Schema

This schema supports the evaluation of key performance indicators (KPIs) such as OEE, machine utilization, production efficiency, quality performance and production capacity across different product divisions, machine type and time periods.